Local scalar insertions in dS_3: from normalized BD modes to global CG coefficients

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Abstract

For an explicitly chosen $G = 0$ local multiplication insertion, we compute the scalar trilinear coefficient using Klein–Gordon normalized Bunch–Davies modes in global dS_3. A finite Beta sum evaluates arbitrary finite angular-mode entries. Future boundary data fix the unitary shadow phases, and a normalized spherical period fixes the full complex global Clebsch–Gordan coefficient. Its norm differs from the previously extracted paired C_0 factor by explicit leg factors. This is a local dictionary in a declared model, not a construction of the physical observer mass measure or finite-gravity sewing.

1 The vacuum and its normalized modes

Set the radius to one and take

$$ds^2 = -dt^2 + \cosh^2 t\, d\Omega_2^2, \qquad m_{\text{eff}}^2 = 1 + \nu^2, \quad \nu > 0. \tag{1}$$

Use $\int |Y_{lm}|^2 \, \mathrm{d}\Omega = 1$ and $Y_{00} = 1/\sqrt{4\pi}$. The angular momentum l is an SO(3) K-type label. A single scalar principal-series representation, with global spin $J = 0$, contains *every* $l \geq 0$.

For $u_{\nu lm} = f_{\nu l} Y_{lm}$ and $\chi_{\nu l} = \cosh t\, f_{\nu l}$,

$$\chi'' + [\nu^2 + l(l+1)\operatorname{sech}^2 t]\chi = 0, \qquad i(\chi^* \chi' - \chi'^* \chi) = 1. \tag{2}$$

LMW [1] distinguish the reflectionless in/out modes from the Euclidean or BD modes. Regularity on the lower Euclidean hemisphere and KG normalization give

$$\chi_{\nu 0} = \frac{\sinh(\pi\nu/2 - i\nu t)}{\sqrt{\nu \sinh \pi\nu}} = N_\nu (e^{-i\nu t} - q_\nu e^{i\nu t}), \quad q_\nu = e^{-\pi\nu}, \quad N_\nu = [2\nu(1 - q_\nu^2)]^{-1/2}. \tag{3}$$

The numerator vanishes at $t = -i\pi/2$, making $f_{\nu 0}$ regular. Omitting the second exponential changes the vacuum and the vertex.

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2 An actual local insertion and a convergent mode integral

Define $D_j = \partial_t - j\tanh t$ and $d_l(\nu) = \prod_{j=1}^{l}\sqrt{\nu^2+j^2}$. The normalized modes are

$$f_{\nu l} = \frac{i^l}{d_l(\nu)\cosh t} D_l \cdots D_1 \chi_{\nu 0}. \tag{4}$$

Each Darboux step maps the previous equation into (2), multiplies its Wronskian by ν^2+j^2 and preserves the regular Euclidean branch. The phase i^l agrees with MM (16)–(17) [2]:

$$f_{\nu l} = \sqrt{\frac{\Gamma(l+1-i\nu)\Gamma(l+1+i\nu)}{2}}\,(\cosh t)^{-1/2} P^{-l-1/2}_{-1/2+i\nu}(i\sinh t). \tag{5}$$

The type-2 Legendre branch is intended.

Following the structure of KL (5.4) [4], we *define* the present dS_3 model by

$$\langle o|\mathcal{O}_\phi|i\rangle := \int_{\mathrm{dS}_3} \mathrm{d}V\, u_o^*(X)\phi(X)u_i(X). \tag{6}$$

Here $\phi = \sum(a_p u_p + a_p^\dagger u_p^*)$ is a canonically normalized free BD scalar. The local coupling is chosen to be one. Thus the annihilation and emission coefficients are $\int u_o^* u_i u_p\ \mathrm{d}V$ and $\int u_o^* u_i u_p^*\ \mathrm{d}V$. The construction is currently at fixed representation labels; a physical mass direct integral and operator domain are additional tasks.

Write $T_\eta = \int \mathrm{d}V\ \prod_j u_j^{(\eta_j)}$, where $\eta_j = +1$ means u_j and $\eta_j = -1$ means u_j^*. For finite angular modes, $u_j = O(e^{-|t|})$, so the integrand with volume is $O(e^{-|t|})$. The integral converges absolutely. Killing derivatives give a total divergence with vanishing boundary flux, establishing invariance. Finite group translates retain this decay because the induced conformal factors on the compact boundary sphere are bounded.

For explicit evaluation, set $y = \tanh t$ and

$$P_{0,\sigma} = 1, \quad P_{l,\sigma} = (1-y^2)P'_{l-1,\sigma} + (i\sigma\nu - ly)P_{l-1,\sigma}, \qquad \sigma = \pm 1. \tag{7}$$

There are eight exponential branches in a triple product. The saved evaluator uses $m_j = 0$; general magnetic labels enter through the usual conjugated Gaunt coefficient. Every radial term uses

$$I_n(\omega) = \int_{\mathbb{R}} \operatorname{sech} t\ \tanh^n t\, e^{i\omega t}\,\mathrm{d}t = \sum_{k=0}^{n}\binom{n}{k} 2^k(-1)^{n-k} B\left(k+\frac{1+i\omega}{2}, \frac{1-i\omega}{2}\right). \tag{8}$$

This follows by $x = (1+\tanh t)/2$; $I_0 = \pi/\cosh(\pi\omega/2)$. If b_j sums the absolute polynomial coefficients of both normalized branches, truncation to $|t| \le T$ has

$$|E_T| \le 4e^{-T}|\mathcal{G}| \prod_j b_j. \tag{9}$$

This bounds the tail, not the numerical integration error inside it.

3 The spherical entry, future data and shadow phases

For $S = \nu_1 + \nu_2 + \nu_3$, introduce

$$\mathcal{D} = \cosh\frac{\pi S}{2}\prod_j \cosh\frac{\pi(S-2\nu_j)}{2}, \qquad B_* = \frac{\pi}{4\sqrt{4\pi}\mathcal{D}}\sqrt{\prod_j \frac{\sinh \pi\nu_j}{\nu_j}}. \tag{10}$$

Expanding (3) into eight terms and using I_0 gives

Conjugated legs	$T_{\eta,000}$
Zero or three	$-B_*$
One, with conjugated leg r	$B_* \cosh(\pi\nu_r)$
Two, with unconjugated leg r	$B_* \cosh(\pi\nu_r)$

For annihilation r is the outgoing observer; for emission it is the incoming observer. Reversing the observer legs relates the two by Hermitian conjugation. These hyperbolic factors originate in the BD frequency mixture.

Define the unit-modulus coefficients

$$c_l(\nu) = i^l \prod_{j=1}^{l} \frac{-i\nu - j}{\sqrt{\nu^2 + j^2}} = (-i)^l \frac{(1+i\nu)_l}{d_l(\nu)}. \tag{11}$$

Normalized future data for a finite mode combination are $A_\nu = \sum a_{lm} c_l(\nu) Y_{lm}$. This map preserves the KG coefficient norm as an $L^2(S^2)$ norm. The late-time expansion is

$$u(t,n) = 2N_\nu e^{-t}\left[e^{-i\nu t}A_\nu(n) - q_\nu e^{i\nu t}\mathcal{S}_\nu A_\nu(n)\right] + O(e^{-3t}), \tag{12}$$

where

$$\mathcal{S}_\nu Y_{lm} = \frac{(1-i\nu)_l}{(1+i\nu)_l} Y_{lm}, \qquad \mathcal{S}_{-\nu}\mathcal{S}_\nu = 1. \tag{13}$$

The normalized spherical shadow kernel is $i\nu(\pi 4^{i\nu})^{-1}|n-n'|^{-2+2i\nu}$, defined by distributional analytic continuation. Its ordinary integral is not absolutely convergent at these parameters. The two future branches are related data in one BD space.

Put $w_j = 1 + i\eta_j\nu_j$ and

$$a_i = -\frac{1}{2} + \frac{i}{2}(\eta_j\nu_j + \eta_k\nu_k - \eta_i\nu_i), \qquad \mathcal{K}_\eta = \prod_{\text{cyc}} |n_j - n_k|^{2a_i}. \tag{14}$$

The trilinear kernel has weights $2 - w_j$, not w_j. Under stereographic projection with $\Omega(z) = 2/(1+|z|^2)$, $\mathrm{d}\Omega_2 = \Omega^2\,\mathrm{d}^2 z$, $|n_i - n_j|^2 = \Omega_i\Omega_j|z_i - z_j|^2$, and $A_{\text{plane}} = \Omega^w A_{\text{sphere}}$. All factors cancel since $2 - w_i + a_j + a_k = 0$. For annihilation the resulting plane kernel is exactly DS (17) [3], with inputs (i,p), output o, and $a_{\text{DS}} = -i\nu$. Emission uses the conjugate reverse kernel.

4 The normalized spherical period fixes the coefficient

Let $A = a_1 + a_2 + a_3$ and normalize the spherical average by

$$J(a_1, a_2, a_3) = \frac{1}{(4\pi)^3} \int (\mathrm{d}\Omega)^3 \prod_{\mathrm{cyc}} |n_j - n_k|^{2a_i} = 4^A \frac{\Gamma(A+2) \prod_i \Gamma(a_i+1)}{\prod_i \Gamma(A - a_i + 2)}. \tag{15}$$

The domain $\operatorname{Re} a_i > -1$, $\operatorname{Re} A > -2$ ensures absolute convergence; our principal labels have $\operatorname{Re} a_i = -1/2$, $\operatorname{Re} A = -3/2$.

For an independent normalization check, the spherical convolution with $(2-2x)^a$ has eigenvalue

$$\lambda_l(a) = \frac{4\pi\, 4^a}{a+1} \frac{(-a)_l}{(a+2)_l}. \tag{16}$$

Taking the trace of three such convolutions reduces J to

$$\frac{4^A}{\prod_i (a_i+1)} \sum_{l=0}^{\infty} (2l+1) \prod_i \frac{(-a_i)_l}{(a_i+2)_l}. \tag{17}$$

The Rogers–Dougall sum, DLMF 16.4.9 [6], yields (15). Establish the trace identity for smooth kernels first and continue analytically to the convergence domain. The summand is $O(l^{-2\operatorname{Re} A - 5})$, hence $O(l^{-2})$ here. Direct checks give $J(0,0,0) = 1$ and $J(a,0,0) = 4^a/(a+1)$. This rederives a standard triangular integral [5], not a new summation identity.

Define $\mathcal{B}_\eta = \int \prod_j \mathrm{d}\Omega_j\, A_j^{(\eta_j)} \mathcal{K}_\eta$. For generic principal-series labels the standard conformal three-point covariance fixes this kernel up to one overall coefficient, as in DS (17)–(19). The actual integral fixes that coefficient without fitting:

$$\boxed{T_\eta = C_\eta \mathcal{B}_\eta, \qquad C_\eta = \frac{T_{\eta,000}}{(4\pi)^{3/2} J(a_1, a_2, a_3)}.} \tag{18}$$

All normalized $l = 0$ boundary data are Y_{00}, explaining the factor $(4\pi)^{3/2}$. Equation (18) retains the complex phase. Gamma reflection simplifies its norm to

$$|C_\eta|^2 = \frac{F_\eta}{64\pi^3 \mathcal{D}}, \qquad F_\eta = \begin{cases} 1, & \text{zero or three conjugated legs,} \\ \cosh^2(\pi\nu_r), & \text{one or two conjugated legs, } r \text{ as above.} \end{cases} \tag{19}$$

For the nontrivial $(l, l, 0)$ entries, rotational averaging produces $P_l(n_1 \cdot n_2)$; expanding $P_l(1 - |n_1 - n_2|^2/2)$ gives a finite sum of (15) with integer shifts of a_3. The two c_l phases must also be included. Independent bulk Beta sums match these boundary entries at $l = 1, 2, 3$, and separate boost Ward tests check higher angular couplings. These tests support the normalization and conjugation conventions, not an extension to a complete physical operator algebra.

5 Comparison with the paired vertex and its remaining scope

The frozen fixed-mass paired C_0 factor extracted from CLP [7], up to its common convention-dependent constant, is

$$D_{\mathrm{CLP}} = \frac{\pi e^{\pi S}}{\mathcal{D}} \prod_j \frac{\sinh \pi\nu_j}{\nu_j}. \tag{20}$$

Equations (18)–(19) imply

$$\frac{|C_\eta|^2}{D_{\mathrm{CLP}}} = \frac{F_\eta}{64\pi^4} \prod_j \frac{\nu_j e^{-\pi\nu_j}}{\sinh \pi\nu_j}. \tag{21}$$

The ratio consists of explicit leg factors, with the distinguished $\cosh^2$ factor fixed by the annihilation/emission orientation. This provides a local $G = 0$ normalization interface. It does not repair the earlier normalization-independent mismatch between the recoil graph H (or Z_{rec}) and the proposed global $6j$.

Nor can (21) simply be substituted for the unknown two-point insertion n. That spectrum also involved the observer weight z, a mass integration protocol, state preparation and sewing norms. Choosing (6) fixes a coupling in this model; symmetry alone would still allow mass-dependent couplings in other models.

Restoring the radius gives each KG-normalized scalar mode a factor $\ell^{-1/2}$ and the three-mode volume integral a factor $\ell^{3/2}$. A physical observable additionally requires the dimensions and normalization of its chosen local coupling. No finite-G four-point dynamics is inferred from this free-field calculation.

The immediate next task is to distinguish fixed-mass KG normalization from spacetime L^2 or mass-delta normalization of an observer. The global radial spectrum has two continuum branches; its discrete sector and any imposed mass threshold must also be identified before projecting to a BD observer space. Only then should the coefficient, DS reflection term and spectral measure be assembled into a physical two-point object.

The independent free-probe, large-observer-mass benchmark remains

$$g_\nu(0) g_{\nu'}(0) = \frac{\nu\nu'}{16\pi^2 \sinh \pi\nu \, \sinh \pi\nu'}. \tag{22}$$

It is a disconnected antipodal product, not a zero-energy graph value. Defining an actual mass-space operator, its group-averaged domain, physical four-point sewing and finite-G corrections remains necessary for the full holographic objective.

6 Verification and source cautions

The saved mode/vertex diagnostics contain 249 passed checks: 108 KG, ODE and independent Legendre checks; 48 spherical-entry and coefficient-norm checks; 27 higher-angular boundary matches; 27 boost Ward checks; eight direct-integration/refinement checks; six spherical-period checks; 16 shadow checks; and nine precision comparisons. A separate record contains 24 checks of (21). These counts are implementation checks, not independent physical predictions.

Mass triples were $(.7, 1.1, .6)$, $(1.4, .9, 1.2)$ and $(3.2, 2.7, .5)$. Calculations used 60 digits, with selected nontrivial entries repeated at 40 digits. The maximum recorded scaled error was 6.34×10^{-38}. Four direct time integrals used $T = 96$, with subdivisions of width three and two; the largest relative difference from the Beta result was 1.55×10^{-41}. The corresponding analytic tail-to-signal bound was at most 2.46×10^{-40}. The largest higher-angular cancellation ratio was 632.41. Interior quadratures are not interval-certified. Python 3.14.5, mpmath 1.3.0, and SymPy 1.14.0, along with full complex values and source hashes, are saved.

Two source inconsistencies are explicitly isolated. MM HTML (14) prints a $\cosh^2 t$ angular coefficient; the metric (4), LMW (3) and the normalized modes instead require $\operatorname{sech}^2 t$. We derived the equation from the metric. Also, combining BEGMP HTML (93), (127), (128) at $d = 3$ gives $J(0, 0, 0) = 16$, inconsistent with their normalized definition. The same constant appears in the cached PDF's (74), (101), (102). We use the independent derivation (15), with its zero- and single-edge checks. Neither observation is presented as an author-confirmed erratum; source files remain unchanged.

References

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